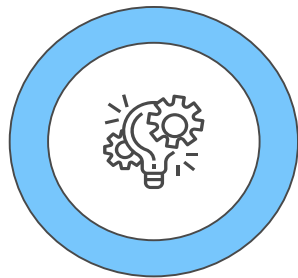


What is Cloud Computing?





Azure Cloud Computing

- Azure is Cloud Computing Platform

- **Offer SERVICES**
- Compute services
- Storage services
- Database services
- Software services

...

Consumption-based pricing

"Pay-as-you-go"



Computing



Storage

01

Build own data center
Server, infrastructure, staff etc.

...

02

Rent service from cloud provider
Compute, storage, etc.

...

Azure Cloud

CapEx vs. OpEx

01

...

Capital Expenditure (CapEx)

High upfront cost,
Own infrastructure, hardware cost
Expenses will be deducted over time

Examples: Buying server

Own data center



02

...

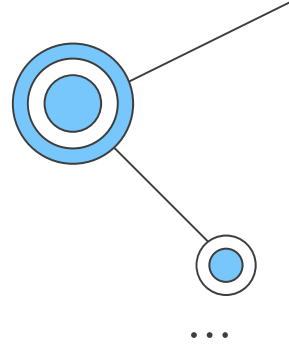
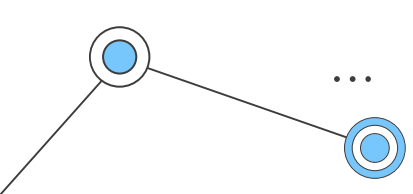
Operational Expenditure (OpEx)

No upfront cost,
Product or services, that can be paid when used ("rented")
Expenses can be immediately deducted

Examples: Azure Cloud,
Monthly payments (employees, electricity, software licenses)

Azure "Pay-as-you-go"





Benefits of the cloud

01

High Availability

Continuous functioning of services

- Access to services for a high percentage of time
- E.g. 99.9% availability = 0.1% downtime

02

Fault tolerance

Ability to respond to failure of a component

- Ability to maintain functionality when components of a system break down
- Minimizing effects from failures of parts of a system

03

Disaster recovery

Ability to restore functionality after a disaster or catastrophe

- Backup and recovery strategies

Benefits of the cloud

04

Scalability

Ability to handle increased load

- Vertical scalability:  Scale up *More CPU per VM*
- Horizontal scalability:  Scale out *More virtual machines*

05

Elasticity

Ability to quickly react to changing demands

- Dynamic scalability
- More users ⇒ automatically scale up and down

06

Agility

Ability to quickly deploy and configure solutions when requirements change

07

Geo-Distribution

Distribute service globally close to users

- Insuring performance globally by regional deployment

Cloud models

Public Cloud

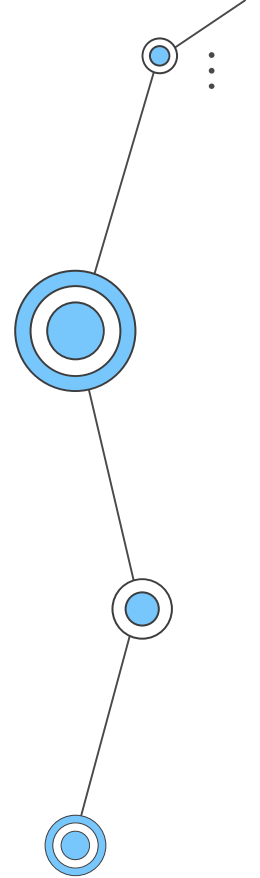
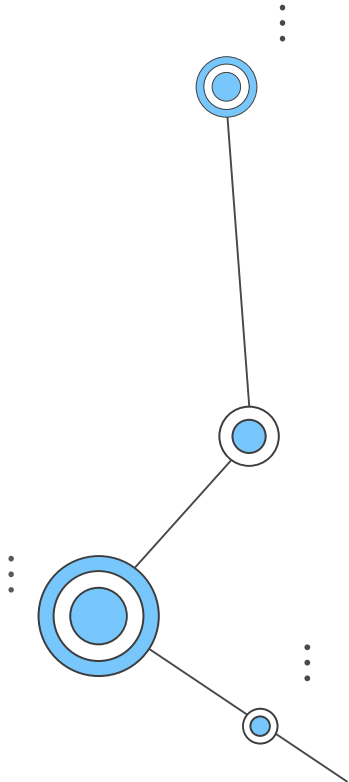
Azure Cloud – Most common model

Private Cloud

Own "cloud" / own data center

Hybrid Cloud

Combination of both



Cloud models

Public Cloud

Azure Cloud – Most common model

Shared hardware that everyone can use

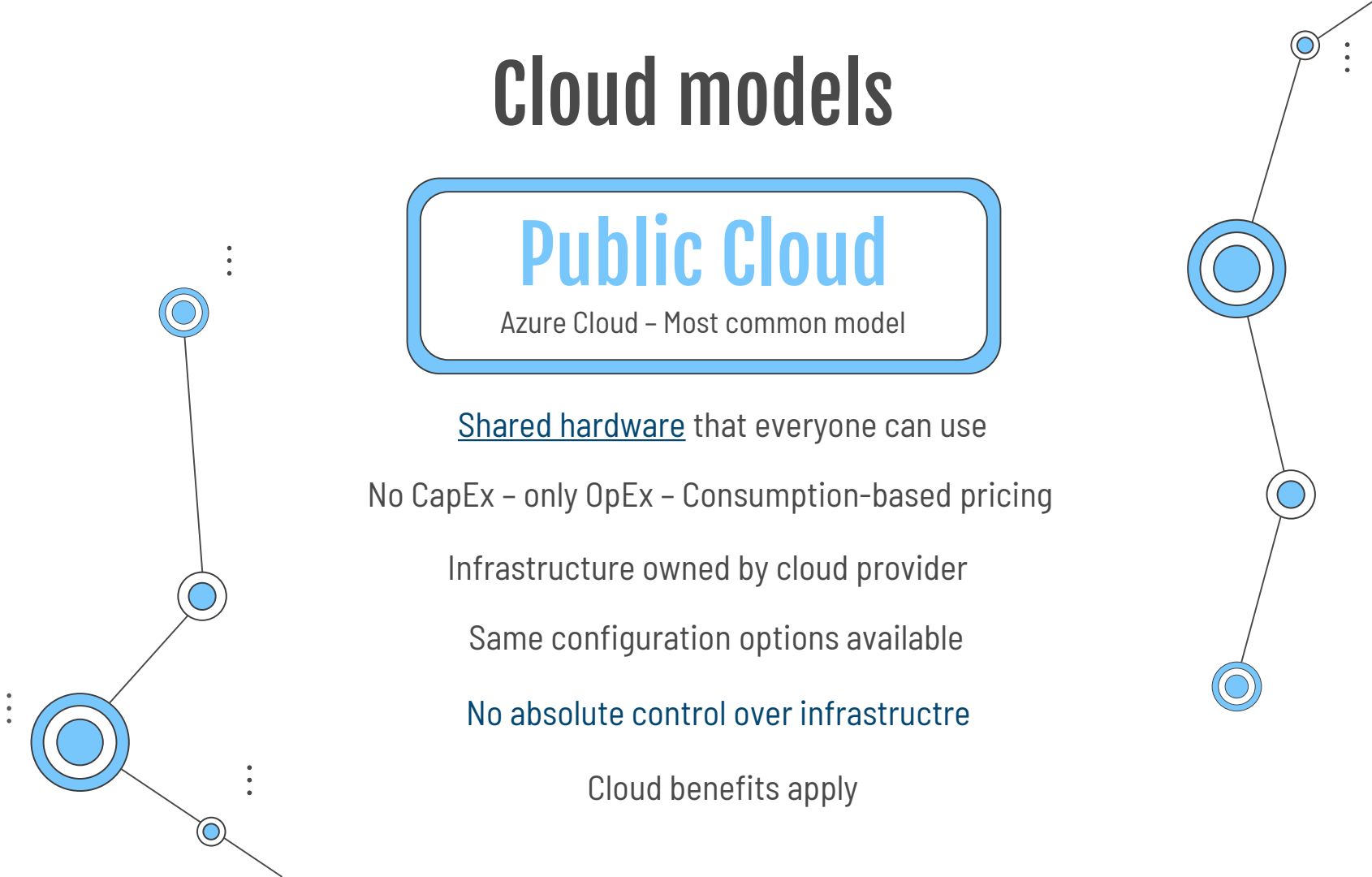
No CapEx – only OpEx – Consumption-based pricing

Infrastructure owned by cloud provider

Same configuration options available

No absolute control over infrastructure

Cloud benefits apply



Cloud models

Private Cloud

Private data center

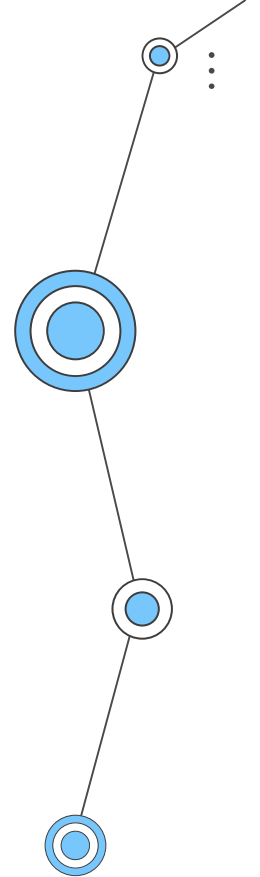
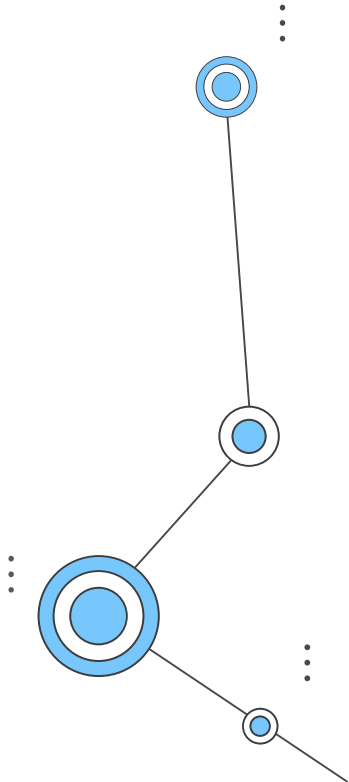
In special situations (e.g. legal / compliance reasons)

Hardware is owned by the company

Control all aspects of hardware

Capital Expenditure (CapEx) applies

Complete responsibility for hardware, security etc.



Cloud models

Hybrid Cloud

Combination of Public & Private

Some (critical) resources are owned
others are used from the Public cloud

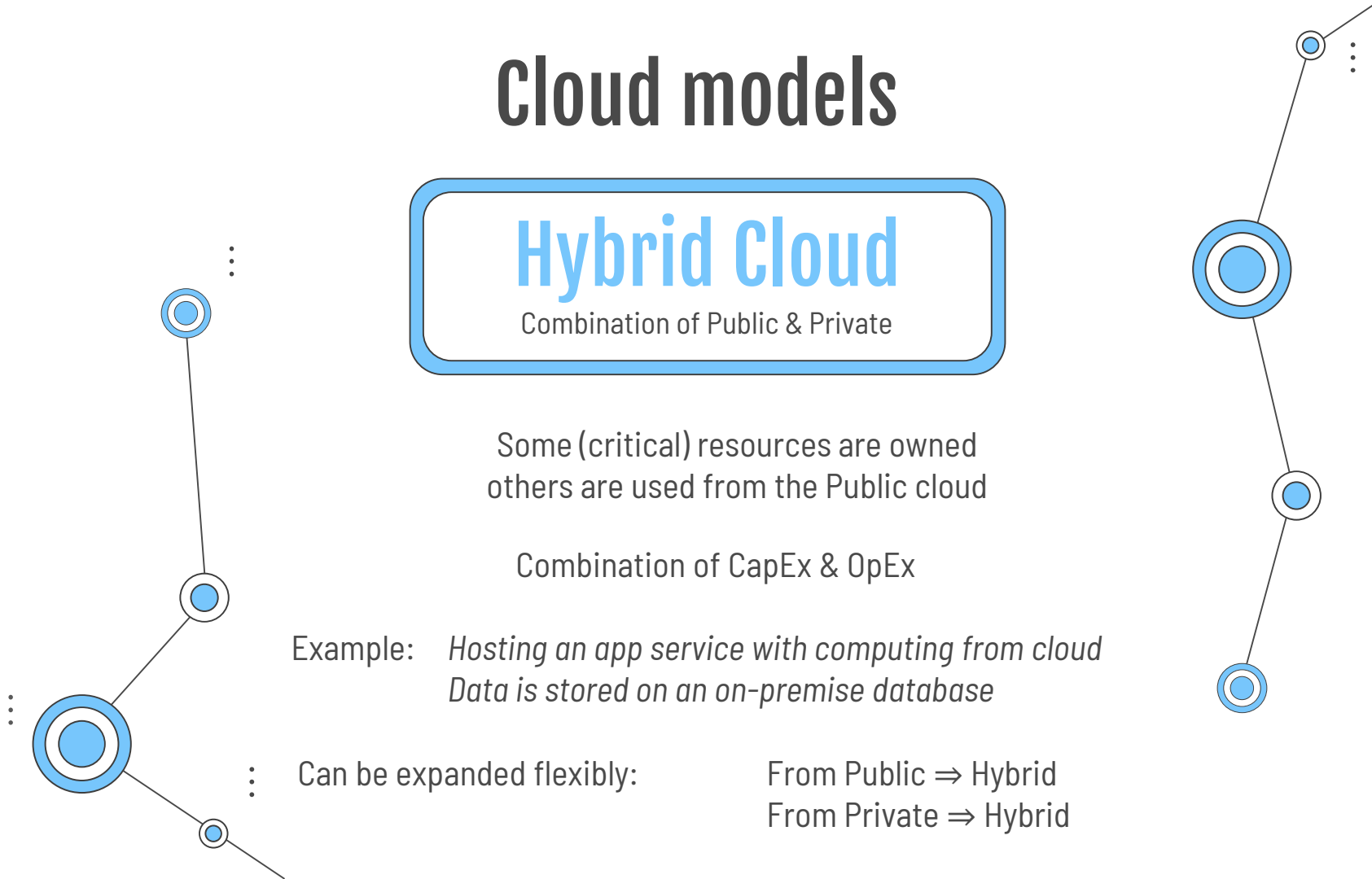
Combination of CapEx & OpEx

Example: *Hosting an app service with computing from cloud
Data is stored on an on-premise database*

Can be expanded flexibly:

From Public $\Rightarrow$ Hybrid

From Private $\Rightarrow$ Hybrid



Cloud service types

IaaS

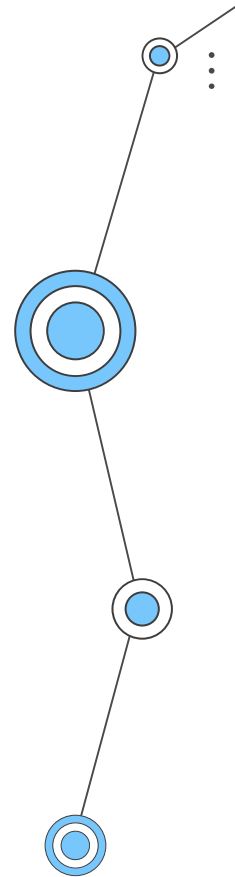
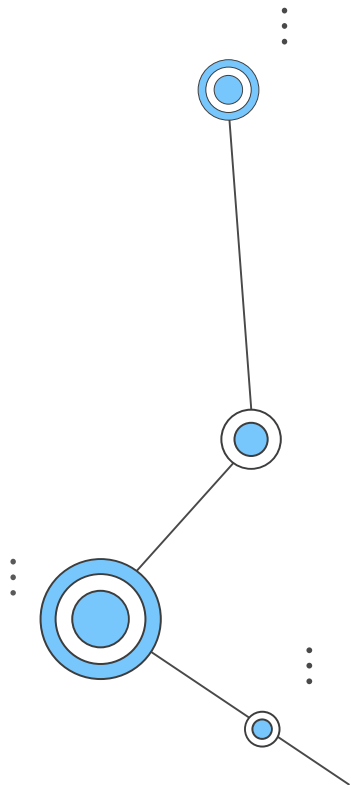
Infrastructure-as-a-Service

PaaS

Platform-as-a-Service

SaaS

Software-as-a-Service



Cloud Services

IaaS

Infrastructure-as-a-Service

Renting of hardware / infrastructure (servers, storage, etc.)

Virtual machines

Storage
(BLOB storage, Azure Files)

Networking
(Virtual Networks, Load Balancers etc.)

Most flexible type of services

Maximum control / special custom applications

Highest responsibility

Pay-as-you-go

 CUSTOMER

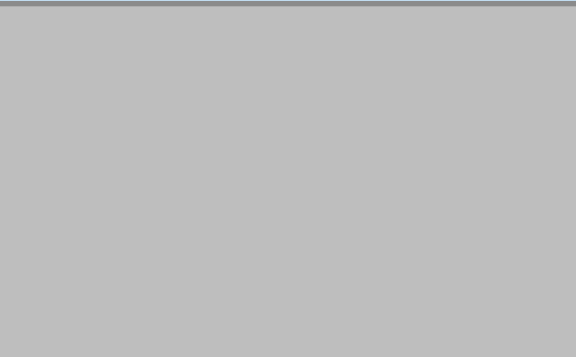
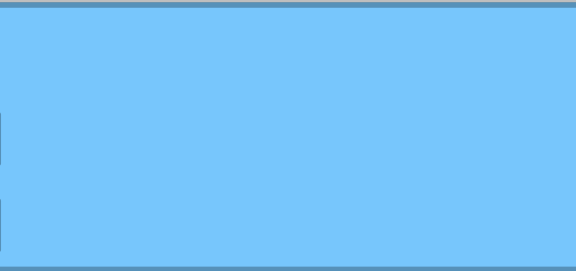
 CLOUD PROVIDER

Responsibilities

Responsibility always with
CUSTOMER

Responsibility
VARIES

Responsibility transfers to
CLOUD PROVIDER

Information and data		
Devices		
Accounts & identities		
Identity & directory infrastructure		
Applications		
Network controls		
Operating systems		
Physical hosts		
Physical network		
Physical data center		

Cloud Services

PaaS

Platform-as-a-Service

Hardware fully managed by Azure

No hardware configuration required

Reduce the administrative effort

Azure SQL database

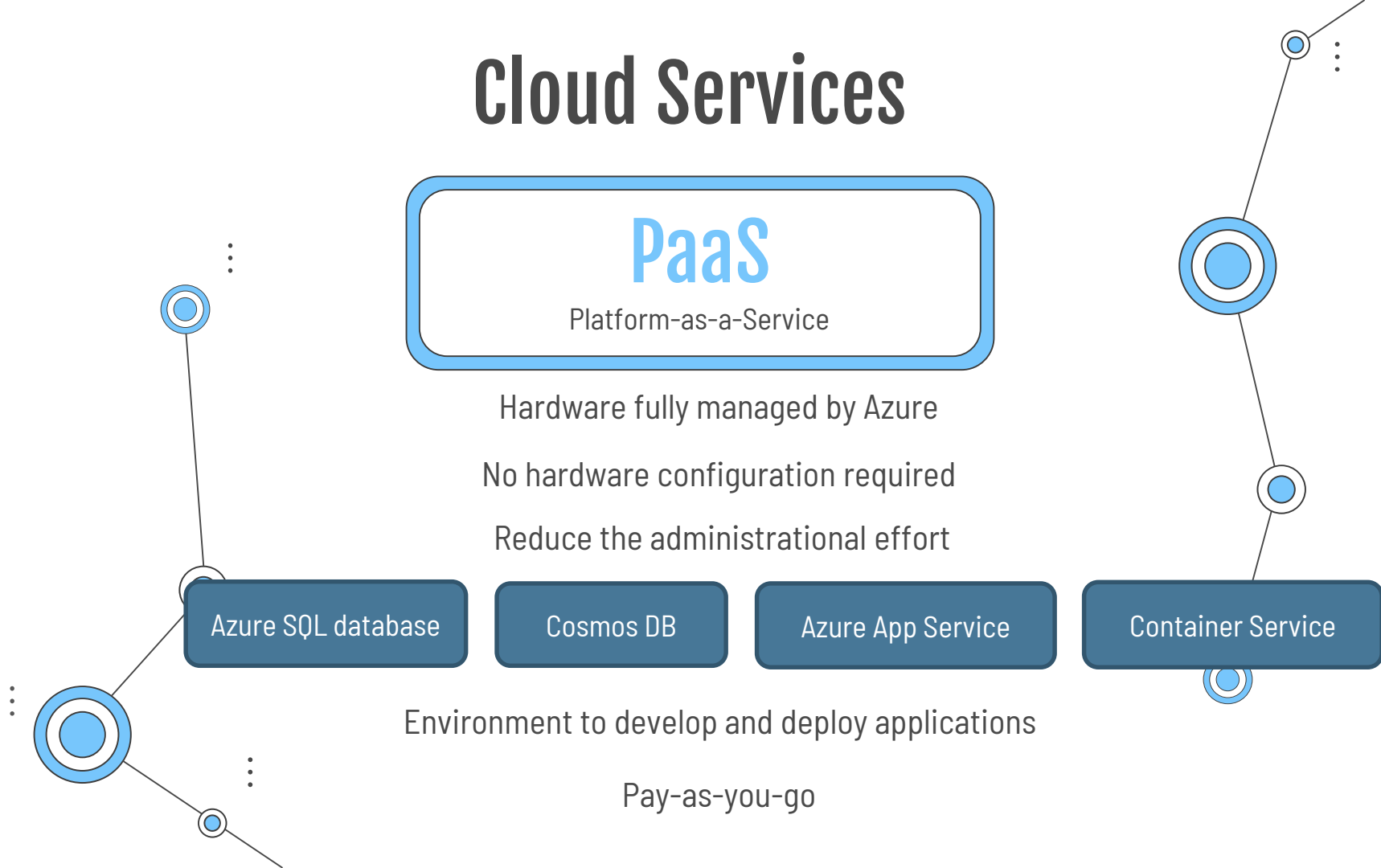
Cosmos DB

Azure App Service

Container Service

Environment to develop and deploy applications

Pay-as-you-go



 CUSTOMER

 CLOUD PROVIDER

Responsibilities

IaaS

PaaS

Responsibility always with
CUSTOMER

Responsibility
VARIES

Responsibility transfers to
CLOUD PROVIDER

Information and data		
Devices		
Accounts & identities		
Identity & directory infrastructure		
Applications		
Network controls		
Operating systems		
Physical hosts		
Physical network		
Physical data center		

Cloud Services

SaaS

Software-as-a-Service

Only the end application is used

Microsoft Office
365

Outlook

Microsoft Planner

No installation needed

Application is not developed – but only configured

Infrastructure remains invisible to customers

Pay-as-you-go model

 CUSTOMER

 CLOUD PROVIDER

Responsibilities

IaaS

PaaS

SaaS

Responsibility always with
CUSTOMER

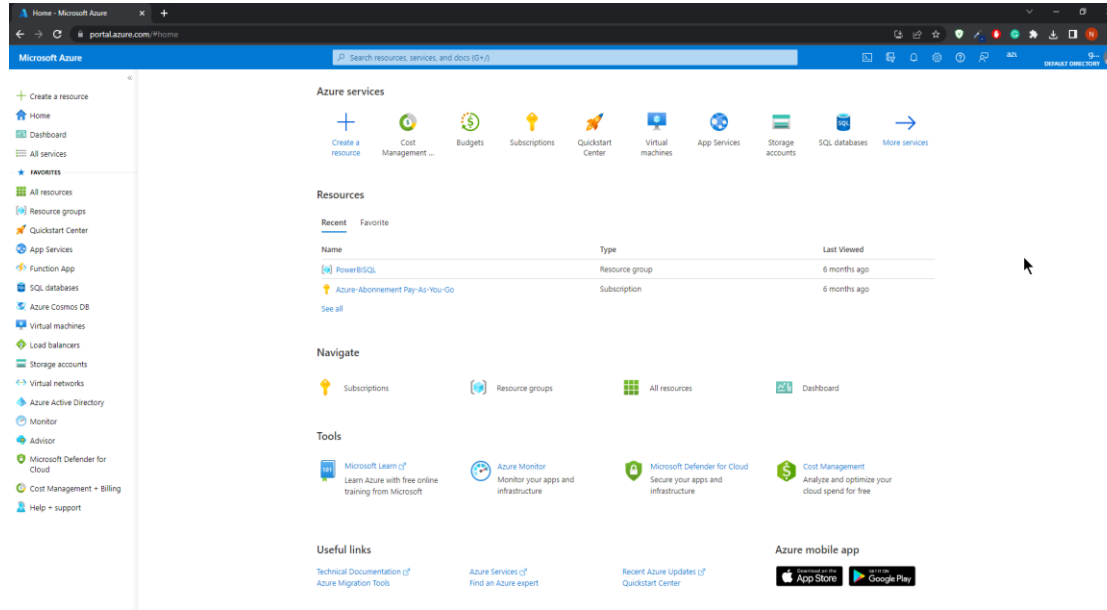
Responsibility
VARIES

Responsibility transfers to
CLOUD PROVIDER

Information and data			
Devices			
Accounts & identities			
Identity & directory infrastructure			
Applications			
Network controls			
Operating systems			
Physical hosts			
Physical network			
Physical data center			

Azure Portal

Browser-based graphical user interface



Can be accessed with any device that has a browser

Consumption-based model

- ✓ Pay only for what is used.

Storage



\$0.01 per GB / per month

Functions



Per execution

⇒ Pay per minute/day/execution/operation/volume

- ✓ No upfront investment / infrastructure necessary.
- ✓ Resources that are no longer needed, no longer need to be paid.

Summary

✓ **Cloud Computing:** Services through the internet

✓ **Benefits of the cloud:**

High Availability

Continuous functioning of services

Fault tolerance

Ability to respond to failure of a component

Disaster recovery

Ability to restore functionality after a disaster or catastrophe

Scalability

Ability to handle increased load

Elasticity

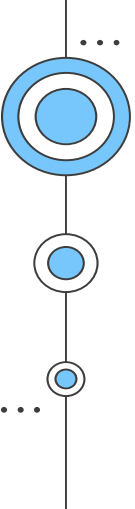
Ability to quickly react to changing demands

Agility

Ability to quickly deploy and configure solutions when requirements change

Geo-Distribution

Distribute service globally close to users



Summary

✓ **CapEx:** Upfront cost & own infrastructure

OpEx: No upfront cost & "pay-as-you-go"

✓ **Public Cloud:** Shared hardware, services from the internet

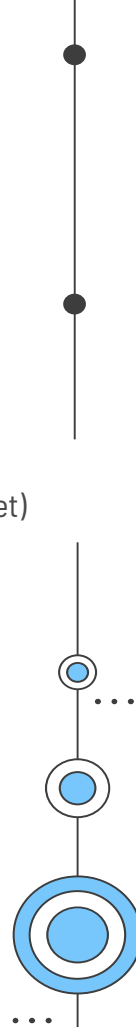
Private Cloud: Own private data center, absolute control (can be connected to internet)

Hybrid Cloud: Combination of both, can come from public & private

✓ **IaaS:** Most control, VMs, Storage, Networking like VNets

PaaS: Mostly managed, less administrative effort, Databases, App Service

SaaS: Use only the end product, no installation, only configuring, email provider, Office 365 etc.



Summary

		On-Prem	IaaS	PaaS	SaaS
Responsibility always with CUSTOMER	Information and data	CUSTOMER	CUSTOMER	CUSTOMER	CUSTOMER
	Devices	CUSTOMER	CUSTOMER	CUSTOMER	CUSTOMER
	Accounts & identities	CUSTOMER	CUSTOMER	CUSTOMER	CUSTOMER
Responsibility VARIES	Identity & directory infrastructure	CUSTOMER	CUSTOMER	SPLIT	CLOUD PROVIDER
	Applications	CUSTOMER	CUSTOMER	SPLIT	CLOUD PROVIDER
	Network controls	CUSTOMER	CUSTOMER	SPLIT	CLOUD PROVIDER
	Operating systems	CUSTOMER	CUSTOMER	CLOUD PROVIDER	CLOUD PROVIDER
Responsibility transfers to CLOUD PROVIDER	Physical hosts	CUSTOMER	CLOUD PROVIDER	CLOUD PROVIDER	CLOUD PROVIDER
	Physical network	CUSTOMER	CLOUD PROVIDER	CLOUD PROVIDER	CLOUD PROVIDER
	Physical data center	CUSTOMER	CLOUD PROVIDER	CLOUD PROVIDER	CLOUD PROVIDER

- ✓ **Consumption-based model:** No upfront cost, pay only for what you use